Polar Coordinates & Polar Graphs

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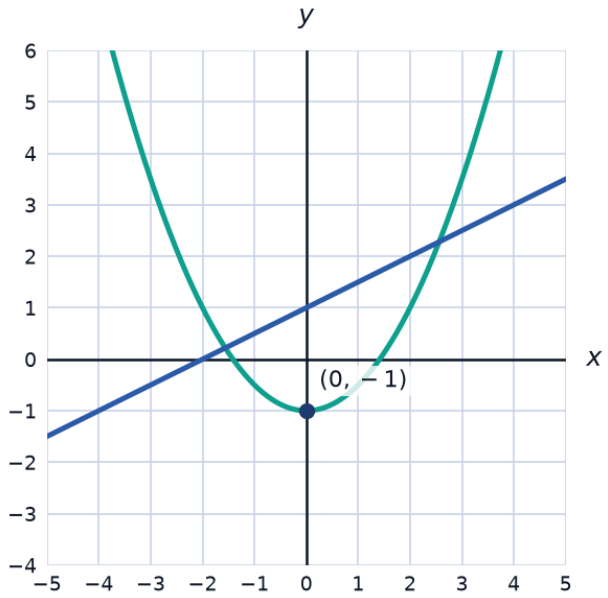
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Coordinate Conversions (Polar <-> Rectangular)**

Convert the rectangular point $(-1, -1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

2 **[Foundational]** • **Target: Converting Equations between Polar and Cartesian**

Convert the rectangular point $(-1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).



3 **[Foundational]** • **Target: Rose Curves and Symmetry**

Convert the rectangular point $(1, -1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

4 **[Foundational]** • **Target: Limaçons and Special Polar Curves**

Convert the rectangular point $(1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

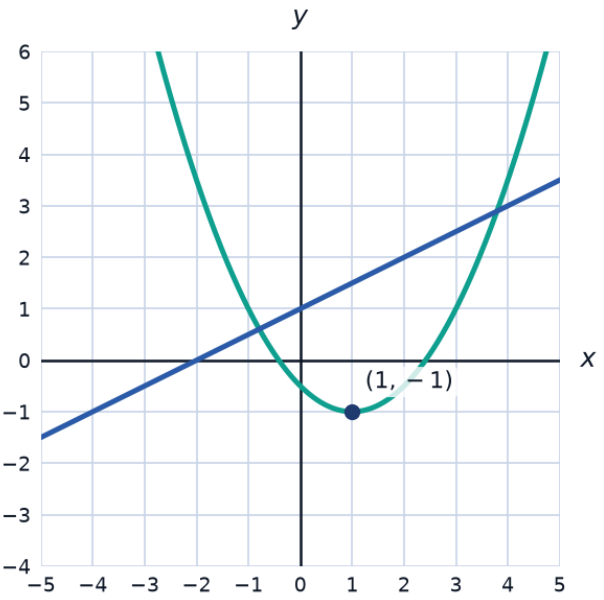
5 **[Foundational]** • **Target: Coordinate Conversions (Polar <-> Rectangular)**

Convert the rectangular point $(-3, 0)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

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6 [Foundational] • Target: Converting Equations between Polar and Cartesian

Convert the rectangular point $(0, -3)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).



7 [Foundational] • Target: Rose Curves and Symmetry

Convert the rectangular point $(0, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limaçons and Special Polar Curves

Convert the rectangular point $(0, 2)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

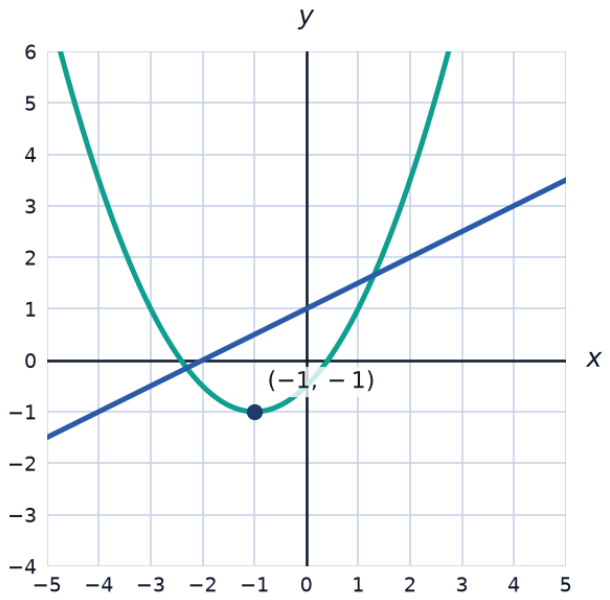
9 [Intermediate] • Target: Coordinate Conversions (Polar <-> Rectangular)

Convert the rectangular point $(2, 0)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

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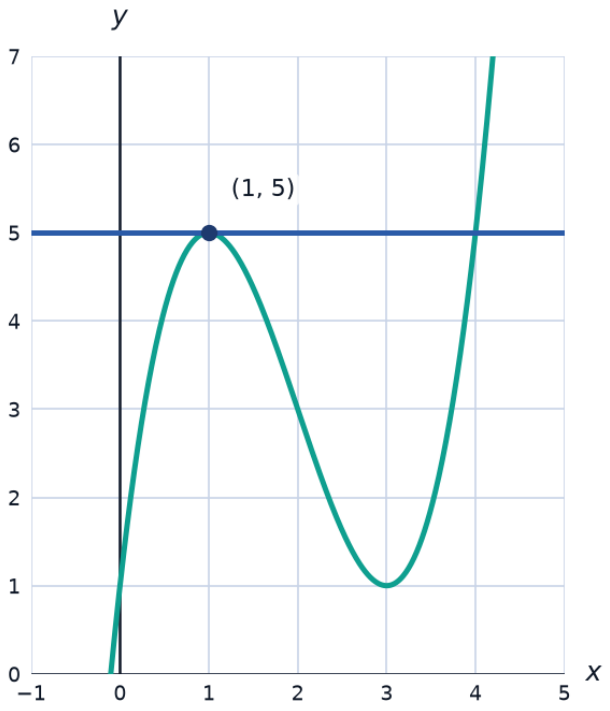
10 **[Intermediate]** • **Target: Converting Equations between Polar and Cartesian**

Convert the rectangular point $(1, 0)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).



11 **[Intermediate]** • **Target: Rose Curves and Symmetry**

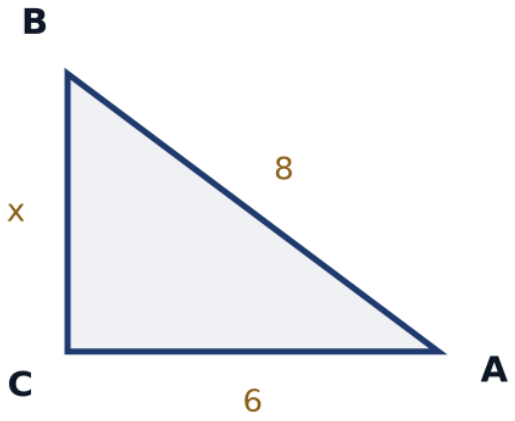
Consider the function $f(x) = x^3 - 6x^2 + 9x + 1$. A line tangent to the graph of $f(x)$ at $x = 1$ intersects the graph at another point P. Determine the x-coordinate of point P.



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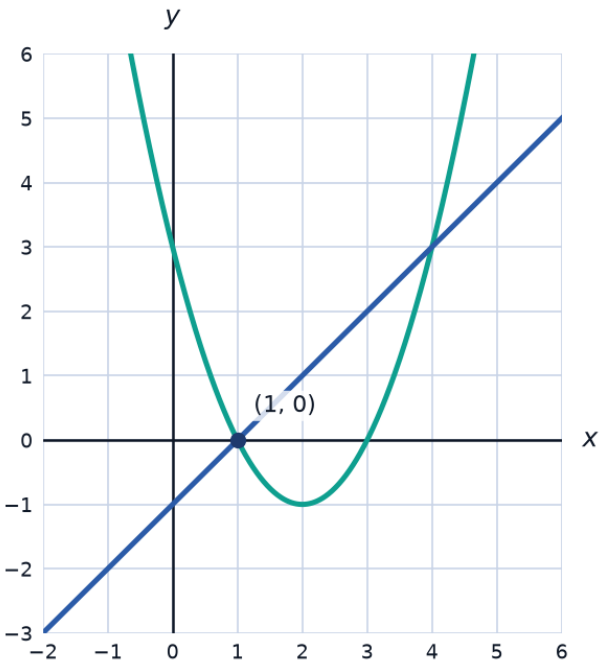
12 [Intermediate] • Target: Limaçons and Special Polar Curves

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$? Show your complete algebraic work.



13 [Intermediate] • Target: Coordinate Conversions (Polar <-> Rectangular)

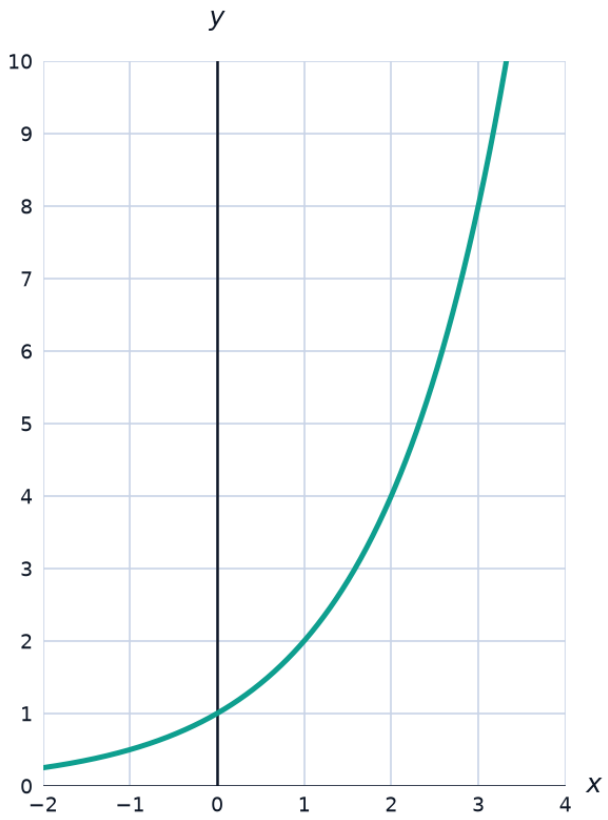
The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at two points. What is the y-coordinate of the vertex of this parabola? Show your complete algebraic work.



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14 [Intermediate] • Target: Converting Equations between Polar and Cartesian

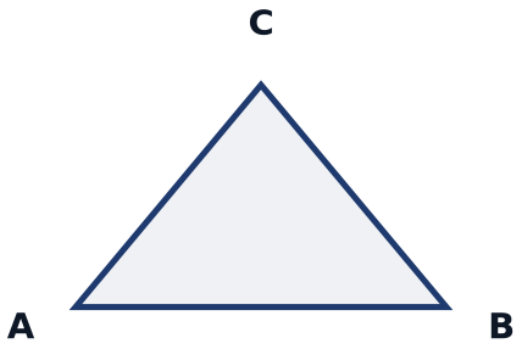
The graph of the exponential function $f(x) = 2^x$ is given. A point P on the graph has a y-coordinate of 8. What is the x-coordinate of point P? Show your complete algebraic work.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 1.03 Rates of Change & Concavity

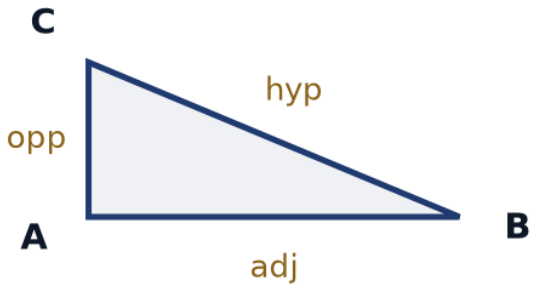
$\tan\theta = \frac{\text{opp}}{\text{adj}}$. In 3-4-5 triangle with opp=3,adj=4:.



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16 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Period of $y = \sin(2x)$ is:.



17 [Challenging] • Target: Coordinate Conversions (Polar <-> Rectangular) • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Consider a right-angled triangle ABC, where the right angle is at vertex B. Point D lies on the hypotenuse AC such that BD is perpendicular to AC. If AB = 10 units and BC = 24 units, find the length of AD.

18 [Challenging] • Target: Converting Equations between Polar and Cartesian • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Consider the exponential function $f(x) = a \cdot b^x$ where a and b are positive constants. The graph of $y = f(x)$ passes through the points $P = (1, 6)$ and $Q = (3, 54)$. A related function $g(x)$ is defined such that $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. If the point R on the graph of $y = g(x)$ has an x-coordinate of 2, what is the y-coordinate of R? Show your complete algebraic work.

19 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 4.01 Inverse & Composite Functions

The terminal ray of an angle θ in standard position in the xy-plane passes through the point $(-5, 12)$. What is the value of $\sin\theta$? Show your complete algebraic work.

20 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

The graph of the function $f(x) = A\sin(Bx + C) + D$ is shown below. The maximum value of the function is 5 and the minimum value is -1. The period of the function is $\frac{\pi}{2}$, and the graph passes through the point $(0, 2)$. Find the value of A.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Coordinate Conversions (Polar <-> Rectangular)

Convert the polar point $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$ to rectangular coordinates (x, y).

a. $(-3\sqrt{3}, 3)$

b. $(3\sqrt{3}, -3)$

c. $(-3, 3\sqrt{3})$

d. $(3, -3\sqrt{3})$

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22 [Foundational] • Target: Coordinate Conversions (Polar <-> Rectangular)

Convert the rectangular point $(-4, 4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $\left(4\sqrt{2}, \frac{3\pi}{4}\right)$

b. $\left(4\sqrt{2}, \frac{7\pi}{4}\right)$

c. $\left(8, \frac{3\pi}{4}\right)$

d. $\left(4\sqrt{2}, \frac{5\pi}{4}\right)$

23 [Foundational] • Target: Coordinate Conversions (Polar <-> Rectangular)

Which polar point represents the same location as $(r, \theta) = \left(3, \frac{\pi}{3}\right)$?

a. $\left(-3, \frac{4\pi}{3}\right)$

b. $\left(-3, \frac{\pi}{3}\right)$

c. $\left(3, -\frac{\pi}{3}\right)$

d. $\left(-3, \frac{2\pi}{3}\right)$

24 [Foundational] • Target: Converting Equations between Polar and Cartesian

Convert the polar equation $r = 6\cos\theta$ into rectangular coordinates.

a. $(x-3)^2 + y^2 = 9$

b. $x^2 + (y-3)^2 = 9$

c. $x^2 + y^2 = 6x$

d. $x^2 + y^2 = 36$

25 [Foundational] • Target: Converting Equations between Polar and Cartesian

Convert the rectangular equation $x^2 + y^2 = 16$ to polar form.

a. $r = 4$

b. $r = 16$

c. $r = 4\cos\theta$

d. $r = 16\sin\theta$

26 [Foundational] • Target: Converting Equations between Polar and Cartesian

Convert $r = 4\csc\theta$ to rectangular coordinates.

a. $y = 4$

b. $x = 4$

c. $y = 4x$

d. $x^2 + y^2 = 16$

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27 [Foundational] • Target: Rose Curves and Symmetry

How many petals does the rose curve $r = 5\cos(4\theta)$ have?

a. 8

b. 4

c. 16

d. 2

28 [Foundational] • Target: Rose Curves and Symmetry

How many petals does the rose curve $r = 3\sin(5\theta)$ have?

a. 5

b. 10

c. 15

d. 3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rose Curves and Symmetry

What is the maximum distance from the pole (origin) to the curve $r = 4\sin(3\theta)$?

a. 4

b. 12

c. 3

d. 8

30 [Intermediate] • Target: Limaçons and Special Polar Curves

Classify the polar curve $r = 2 + 2\cos\theta$.

a. Cardioid (heart-shaped limaçon)

b. Limaçon with an inner loop

c. Dimpled limaçon

d. Convex limaçon

31 [Intermediate] • Target: Limaçons and Special Polar Curves

Which polar equation produces a limaçon with an inner loop?

a. $r = 2 + 4\cos\theta$

b. $r = 4 + 2\cos\theta$

c. $r = 3 + 3\sin\theta$

d. $r = 5 + 4\cos\theta$

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32 [Intermediate] • Target: Limaçons and Special Polar Curves

At what angles $\theta \in [0, 2\pi)$ does the curve $r = 2 + 4\cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$

b. $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

c. $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$

d. $\theta = \pi$

33 [Intermediate] • Target: Coordinate Conversions (Polar <-> Rectangular)

Pythagorean identity: $1 + \tan^2\theta =$

a. $\csc^2\theta$

b. $\sec^2\theta$

c. $\cot^2\theta$

d. $\cos^2\theta$

34 [Intermediate] • Target: Converting Equations between Polar and Cartesian

Law of Cosines: $c^2 =$

a. $a^2 + b^2$

b. $a^2 + b^2 - 2ab\cos C$

c. $a^2 - b^2 + 2ab\cos C$

d. $2ab\cos C$

35 [Intermediate] • Target: Rose Curves and Symmetry

$\sec x =$

a. $\frac{1}{\sin x}$

b. $\frac{1}{\cos x}$

c. $\frac{\sin x}{\cos x}$

d. $\frac{\cos x}{\sin x}$

36 [Intermediate] • Target: Limaçons and Special Polar Curves

For $f(x) = 5\sin(1/2x - 2) + 2$, find the midline.

a. $y = 2$

b. $y = 3$

c. $y = 1$

d. $y = 5$

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37 [Intermediate] • Target: Coordinate Conversions (Polar <-> Rectangular)

For $f(x) = 4\sin(3x - 1) - 1$, find the amplitude.

a. 4

b. 8

c. 5

d. 3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Converting Equations between Polar and Cartesian • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

For $f(x) = 3\sin(x - 1) - 2$, find the amplitude.

a. 2

b. 4

c. 3

d. 6

39 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 4.01 Inverse & Composite Functions

For $f(x) = 3\cos(2x + 2) - 4$, find the amplitude.

a. 4

b. 2

c. 3

d. 6

40 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

For $f(x) = 3\sin(x + 2) - 3$, find the amplitude.

a. 6

b. 3

c. 4

d. 2

41 [Challenging] • Target: Coordinate Conversions (Polar <-> Rectangular) • Cross-Lesson: 4.03 Parametric Equations & Modeling

Evaluate $\arcsin\left(\sqrt{3}/2\right)$.

a. $5\pi/6$

b. $\pi/3$

c. 0

d. $\pi/2$

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42 [Challenging] • Target: Converting Equations between Polar and Cartesian • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Evaluate $\arcsin\left(\sqrt{2}/2\right)$.

a. $3\pi/4$

b. π

c. $\pi/3$

d. $\pi/4$

43 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Evaluate $\arctan\left(\sqrt{3}/3\right)$.

a. 0

b. $\pi/2$

c. $\pi/6$

d. $2\pi/3$

44 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 1.03 Rates of Change & Concavity

Evaluate $\arctan\left(\sqrt{3}\right)$.

a. $5\pi/6$

b. $2\pi/3$

c. $\pi/2$

d. $\pi/3$

45 [Challenging] • Target: Coordinate Conversions (Polar <-> Rectangular) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Identify the polar curve $r = 5(1 + \cos\theta)$.

a. a circle

b. a rose with 4 petals

c. a limaçon with an inner loop

d. a cardioid (heart-shaped curve)

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